ICIC Express Letters
Volume **19**, Number **2**, February **2025**

ICIC International ©2025 ISSN 1881-803X
pp. 231–240

AN INVESTIGATION OF PM2.5 CONCENTRATION PREDICTION WITH AEROSOL OPTICAL DEPTH AND METEOROLOGICAL FACTORS IN SOUTHERN THAILAND USING MACHINE LEARNING TECHNIQUES

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Received April 2024; accepted June 2024

Abstract. *High levels of PM2.5 are an indicator of poor air quality, which impacts overall quality of life and is a pressing environmental and public health issue in urban areas. The aims of this research were (i) to develop suitable predictive models using machine learning techniques; and (ii) to compare their predictive performances. In this study, eleven variables including CO, NO_2, O_3, PM10, wind speed, wind direction, temperature, relative humidity, air pressure, rainfall, and AOD were considered to be the factors for PM2.5. All station data were observed daily from two provinces (Phuket and Narathiwat) over 2019-2022. Two machine learning methods, artificial neural networks (ANN) and extreme gradient boosting (XGBoost), were utilized for the predictive model. The traditional multiple linear regression (MLR) was also considered as a baseline model and their performance was assessed by root mean square error (RMSE) and mean absolute error (MAE) values, as well as the coefficient of determination (R^2). Splitting the data into train and test sets of 70 : 30, it was found that XGBoost outperformed MLR and ANN at Narathiwat station, while MLR performed best at the station in Phuket. Specifically, at the station in Narathiwat, the RMSE, MAE, and R-squared produced by XGBoost were 1.8523, 1.4402, and 80.54%, respectively, while those at the station in Phuket produced by MLR were 2.7202, 2.1722, and 61.20%, respectively. However, it is worth noting that the performance could vary depending on the source of data, location, and time duration.*

Keywords: Predictive model, PM2.5, XGBoost, Artificial neural network, Multiple linear regression

1. **Introduction.** Air pollution is a major concern in both developing and developed countries, with approximately half of the global population experiencing increasing exposure to it [1]. Human activities such as industrialization, automobile use, chemical activities, and agriculture generate PM2.5 [2]. Clear evidence of this was observed during the pandemic lockdown in Wuhan, China, from January 23 to February 22, 2020. During this period, source contributions of PM2.5 from potential geographical regions showed reductions, with mean values ranging from 0.22 to 4.36 $\mu g/m^3$ compared to the same period in 2019 [3]. Natural disasters such as volcanic eruptions and wildfires can also increase PM2.5 levels in the atmosphere [4,5]. Prolonged exposure to elevated PM2.5 levels is linked to various health conditions. For example, long-term exposure to PM2.5 can damage the respiratory system [6], and extended exposure to ambient PM2.5 is associated with an increased risk of developing chronic kidney disease (CKD) [7].

DOI: 10.24507/icicel.19.02.231

Various studies have explored factors influencing PM2.5 levels using different methods. Chu et al. [8] used statistical regression in Pittsburgh and found temperature and humidity interactions explained much of the PM2.5 variation. Wang et al. [9] analyzed data in China (2014-2017) and identified temperature as the most significant factor among five meteorological variables affecting PM2.5 levels. In 2014, researchers in London, Madrid, and Athens used principal component and regression analyses to study the impact of local meteorological parameters on PM2.5 concentrations, finding varying contributions across cities [10]. Additionally, Peng et al. [11] applied extreme gradient boosting (XGBoost) and an artificial neural network (ANN) to data in Hunan Province, showing superior predictive capability with XGBoost. The ANN was also utilized to predict air pollutant discharge by container ships [12]. Multiple linear regression is commonly used both as a predictive model and a baseline for comparison [13,14]. In this paper, we use multiple linear regression as a baseline to evaluate alternative methods.

Southern Thailand faces significant air pollution from construction, industry, transportation, and biomass burning, exacerbated by forest fires in Indonesia. This research analyzed data from two stations in southern Thailand: Phuket and Narathiwat. Located on the Malay Peninsula, southern Thailand lies between the South China Sea to the east and the Andaman Sea to the west. Phuket and Narathiwat were chosen to represent cities along these respective seas. Particularly, since Phuket is a world-renowned tourist destination, gaining an insightful understanding of its environmental conditions is important and worth studying.

2. **Materials and Methods.** All data in this study were obtained from the Pollution Control Department of Thailand except for aerosol optical depth (AOD) that was provided from Himawari-8 satellite. Daily data from 2019 to 2022 (1,461 observations) were collected at two stations in southern Thailand. One station is in Phuket, the largest island in Thailand, located in the Andaman Sea; the other is in Narathiwat, a province in southern Thailand, bordering Malaysia. Figure 1 illustrates the trajectories of all variables for

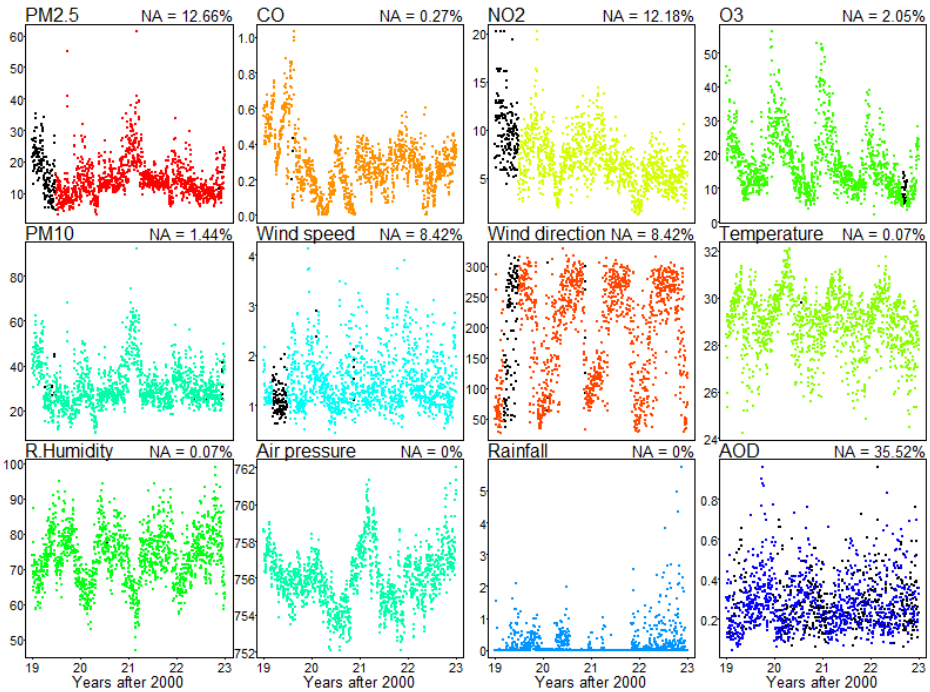

FIGURE 1. Trajectories of the data showing their original values (colored dots) and those imputed (black dots) by the predictive mean matching method for the station in Phuket

Table 1. Descriptive analysis for Narathiwat Provincial Hall

Variables	Min	Max	Mean ± S.D.	Missing values
PM2.5 (μg/m^3)	4.08	84.08	14.94 ± 6.01	85 (5.82%)
CO (ppm)	0.10	1.79	0.84 ± 0.25	846 (57.91%)
NO_2 (ppb)	1.04	14.48	5.17 ± 1.79	102 (6.98%)
O_3 (ppb)	2.78	61.09	20.35 ± 8.17	304 (20.81%)
PM10 (μg/m^3)	9.04	105.54	28.49 ± 9.14	79 (5.41%)
Wind speed (m/s)	0.12	2.22	0.95 ± 0.23	4 (0.27%)
Wind direction (degree)	62.08	294.50	165.05 ± 32.98	4 (0.27%)
Temperature (°C)	22.37	30.66	27.18 ± 1.15	4 (0.27%)
Relative humidity (%)	57.62	98.75	79.08 ± 5.47	4 (0.27%)
Air pressure (mmHg)	751.42	758.75	754.66 ± 1.33	4 (0.27%)
Rainfall (mm)	0.00	20.80	0.34 ± 0.95	4 (0.27%)
AOD	0.06	1.15	0.32 ± 0.18	579 (39.63%)

Table 2. Descriptive analysis for Public Health Center, Phuket

Variables	Min	Max	Mean ± S.D.	Missing values
PM2.5 (μg/m^3)	3.08	68.83	12.17 ± 6.05	185 (12.66%)
CO (ppm)	0.07	1.55	0.79 ± 0.30	4 (0.27%)
NO_2 (ppb)	0	6.91	3.01 ± 0.97	94 (6.43%)
O_3 (ppb)	0	52.26	17.47 ± 7.98	30 (2.05%)
PM10 (μg/m^3)	6.86	87.95	25.05 ± 9.20	21 (1.44%)
Wind speed (m/s)	0.24	3.16	0.96 ± 0.48	123 (8.42%)
Wind direction (degree)	50.54	346.67	244.82 ± 69.23	123 (8.42%)
Temperature (°C)	23.65	40.98	28.41 ± 2.17	1 (0.07%)
Relative humidity (%)	46.33	98.83	77.07 ± 10.21	1 (0.07%)
Air pressure (mmHg)	746.96	774.5	756.3 ± 5.72	0 (0.00%)
Rainfall (mm)	0	4.85	0.19 ± 0.47	0 (0.00%)
AOD	0.06	1.18	0.30 ± 0.14	519 (35.52%)

the Phuket station. Due to their similarity, Narathiwat's trajectories are omitted. Detailed information on the variables' maximum, minimum, and missing values for both stations is presented in Tables 1 and 2.

2.1. **Data gathering and research framework.** All data except for AOD were obtained from the Pollution Control Department, Ministry of Natural Resources and Environment, Thailand.

Himawari-8 is a geostationary satellite sensor system operated by the Japan Meteorological Agency. AOD observations from Himawari-8 were available for both hourly and daily data. However, to maintain consistency with the daily resolution of other ground station variables, daily AOD data was utilized. The pixel size for AOD data captured by Himawari-8 was approximately 0.05×0.05 degrees, equivalent to about 5.5×5.5 square kilometers.

In this study, AOD data from the four closest pixels to each ground station were downloaded and then averaged to represent the AOD at the corresponding station. This approach effectively addressed the issue of missing values in the data.

In accordance with the research framework outlined in Figure 2, data from two sources were integrated, and missing values were imputed by the predictive mean matching (pmm)

method, which is provided by the library MICE in R. Subsequently, the data was divided into training and testing sets in a 7 : 3 ratio, respectively. The training set was then used to train three different models: Multiple Linear Regression, Artificial Neural Network, and Extreme Gradient Boosting. The performance of these models was assessed using metrics such as RMSE (root mean square error), MAE (mean absolute error), and R-squared. In fact, the RMSE and MAE measure the average magnitude of the residuals, while R-squared represents the proportion of variance in the outcome that is explained by the predictors. In other words, lower RMSE and MAE values and a higher R-squared value are preferable for demonstrating better performance of the model. Finally, the final models were applied to the testing set to evaluate their performance on previously unseen data.

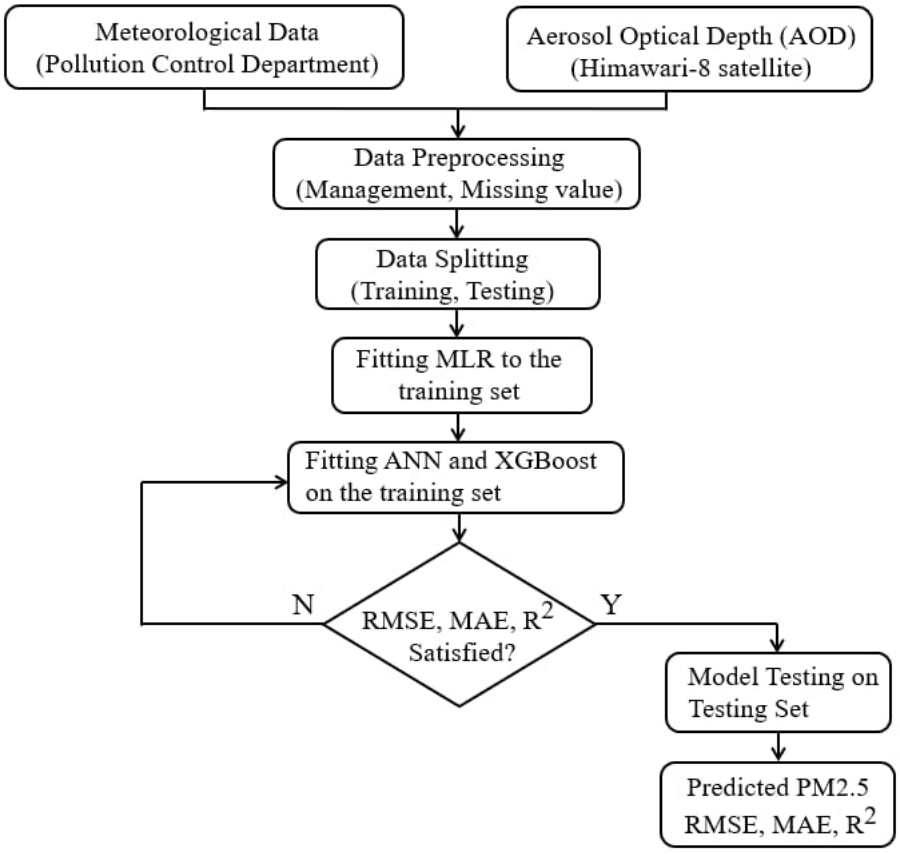

FIGURE 2. Research framework of the study

2.2. **Methods.** Besides preliminary exploration on data, three models were utilized: multiple linear regression (MLR), artificial neural networks (ANN) and extreme gradient boosting. The MLR was applied when there is more than one predictor. Particularly, taking PM2.5 as the outcome variable and the rest are predictors, the MLR took the form [16]:

$$y = \beta_0 + \beta_1 x_1 + \cdots + \beta_n x_n + \varepsilon \tag{1}$$

where y is the outcome variable, $x_1, x_2, \ldots, x_n$ are the predictors (n is 10 in this study), β_0 is the intercept, β_i ($i = 1, 2, 3, \ldots, n$) are the coefficients of the predictors, and ε is the error term.

Artificial neural networks are particularly suitable for capturing non-linear relationships in data. They consist of interconnected nodes (neurons) organized into layers: an input layer, one or more hidden layers, and an output layer. Each neuron in a layer processes

input data and passes it to the next layer through weighted connections. ANNs use activation functions to introduce non-linearity, enabling them to model intricate patterns. Specifically, each neuron in the next layer calculates its value by applying the activation function to the linear combination of the weights, the inputs from the previous layer, and a bias term.

In this study, there are 10 neurons in the input layer, representing 10 predictor variables. An example of a neural network with three neurons in the input layer, three neurons in the hidden layer of a regression problem was illustrated in Figure 3.

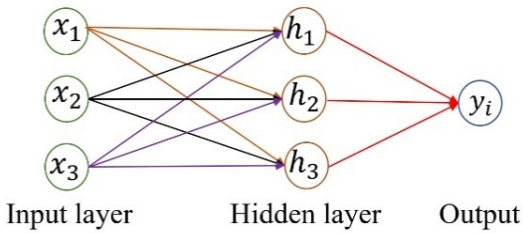

Figure 3. Artificial neural network

Extreme gradient boosting (XGBoost) is an ensemble learning algorithm that combines multiple decision trees to make predictions. It uses gradient boosting to minimize the loss function by adjusting the weights of the trees. XGBoost includes regularization techniques to prevent overfitting. It is known for its high predictive accuracy and can capture complex relationships between the predictor variables and the outcome variable, making it effective for modeling PM2.5 concentration.

3. **Results and Discussion.** The preliminary exploration for maximum and minimum values, the mean ± standard deviation interval, and the count of missing values of both stations were shown in Table 1 (Narathiwat Provincial Hall) and Table 2 (Public Health Center, Phuket). At Narathiwat Provincial Hall (Table 1), CO, AOD, and O_3 had the highest proportions of missing at 57.91%, 39.63%, and 20.81%, respectively, whereas all other variables had missing values of less than 7%. At the public health center in Phuket (Table 2), AOD was missing for over 35%, followed by PM2.5, which had a missing data rate of 12.66%. The remaining variables exhibited missing data rates of less than 9%, with air pressure and rainfall having no missing values. Furthermore, concerning the outcome variable (PM2.5), it was observed that the average (14.94 $\mu g/m^3$) and maximum (84.08 $\mu g/m^3$) values at Narathiwat exceeded those in Phuket, where the average and maximum values were 12.17 $\mu g/m^3$ and 68.83 $\mu g/m^3$, respectively.

Subsequently, predictive models were constructed through the application of multiple linear regression, artificial neural networks, and XGBoost on the training dataset. Utilizing PM2.5 as the dependent variable and the remaining variables as predictors, Tables 3 and 4 present the coefficients of the statistically significant predictors for Narathiwat Provincial Hall and the Public Health Center in Phuket, respectively.

Specifically, in Narathiwat, eight predictors (CO, NO_2, O_3, PM10, wind speed, temperature, air pressure, and rainfall) demonstrated significance, while wind direction, relative humidity, and AOD were not statistically significant (Table 3). Conversely, at the Public Health Center in Phuket, nine predictors (CO, O_3, PM10, wind speed, wind direction, temperature, relative humidity, air pressure, and rainfall) exhibited statistical significance, with the exceptions being NO_2 and AOD (Table 4). The intercepts for the Narathiwat and Phuket stations are 204.95 and −122.10, respectively, interpreting them directly as actual PM2.5 concentrations can be challenging. In a multiple regression model, the intercept usually represents the predicted value of PM2.5 when all predictor variables are set to zero. However, this scenario of zero values for all predictors is rarely, if ever, realistic

Table 3. MLR results for statistically significant factors at the station in Narathiwat

Variables	Estimate	Std. Error	P-value
Intercept	204.95	59.94	< 0.01
CO (ppm)	0.88	0.39	0.02
NO_2 (ppb)	0.15	0.06	0.01
O_3 (ppb)	0.04	0.01	< 0.01
PM10 ($\mu g/m^3$)	0.61	0.01	0.03
Wind speed (m/s)	-2.09	0.50	< 0.01
Temperature (°C)	0.21	0.09	< 0.01
Air pressure (mmHg)	-0.28	0.08	< 0.01
Rainfall (mm)	0.49	0.14	< 0.01

Table 4. MLR results for statistically significant factors at the station in Phuket

Variables	Estimate	Std. Error	P-value
Intercept	-122.10	49.94	0.01
CO (ppm)	-1.78	0.47	< 0.01
O_3 (ppb)	0.11	0.01	< 0.01
PM10 ($\mu g/m^3$)	0.57	0.01	< 0.01
Wind speed (m/s)	-1.15	0.16	< 0.01
Wind direction (degree)	0.00	0.00	0.02
Temperature (°C)	0.31	0.10	< 0.01
Relative humidity (%)	0.15	0.02	< 0.01
Air pressure (mmHg)	0.13	0.06	0.04
Rainfall (mm)	-1.03	0.36	< 0.01

in environmental contexts. Instead, we interpret the intercepts as reference points for understanding the relative influence of each predictor variable within the observed range of values. In other words, the intercept values themselves may not represent achievable concentrations of PM2.5.

After that, the ANN and XGBoost models were trained using the statistically significant predictors identified by the multiple linear regression models at each station. At Narathiwat station, the ANN model utilized eight predictors, including CO, NO_2, O_3, PM10, wind speed, temperature, air pressure, and rainfall. The optimal ANN model, as determined by RMSE, MAE, and R-squared, featured two hidden layers with 5 and 4 neurons, and the number of epochs was 5. In contrast, the ANN model for the Phuket station incorporated nine predictors (CO, O_3, PM10, wind speed, wind direction, temperature, relative humidity, air pressure, and rainfall) and exhibited two hidden layers with 5 and 3 neurons, and the number of epochs was 5. Lastly, the XGBoost model was trained on the same set of predictors as the ANN for each station. Notably, during the training process, it was observed that XGBoost, configured with a maximum tree depth of two, delivered the best performance for both stations.

Once the three suitable models (an MLR, an ANN, and an XGBoost) were obtained, their performances were tested by subjecting the unseen dataset. As a result, XGBoost outperformed the other two models at Narathiwat station with RMSE, MAE, and R^2 of 1.8523, 1.4402, and 80.54%, respectively. On the other hand, at Phuket station, the MLR outperformed the other two models with RMSE, MAE, and R^2 of 2.7202, 2.1722, and 61.20%, respectively.

Moreover, the trajectories of the original PM2.5, its fitted values on the training and testing sets for including their RMSE, MAE, and R-squared values were illustrated in Figures 4 and 5. Specifically, Figure 4 represents the trajectory results using MLR, ANN,

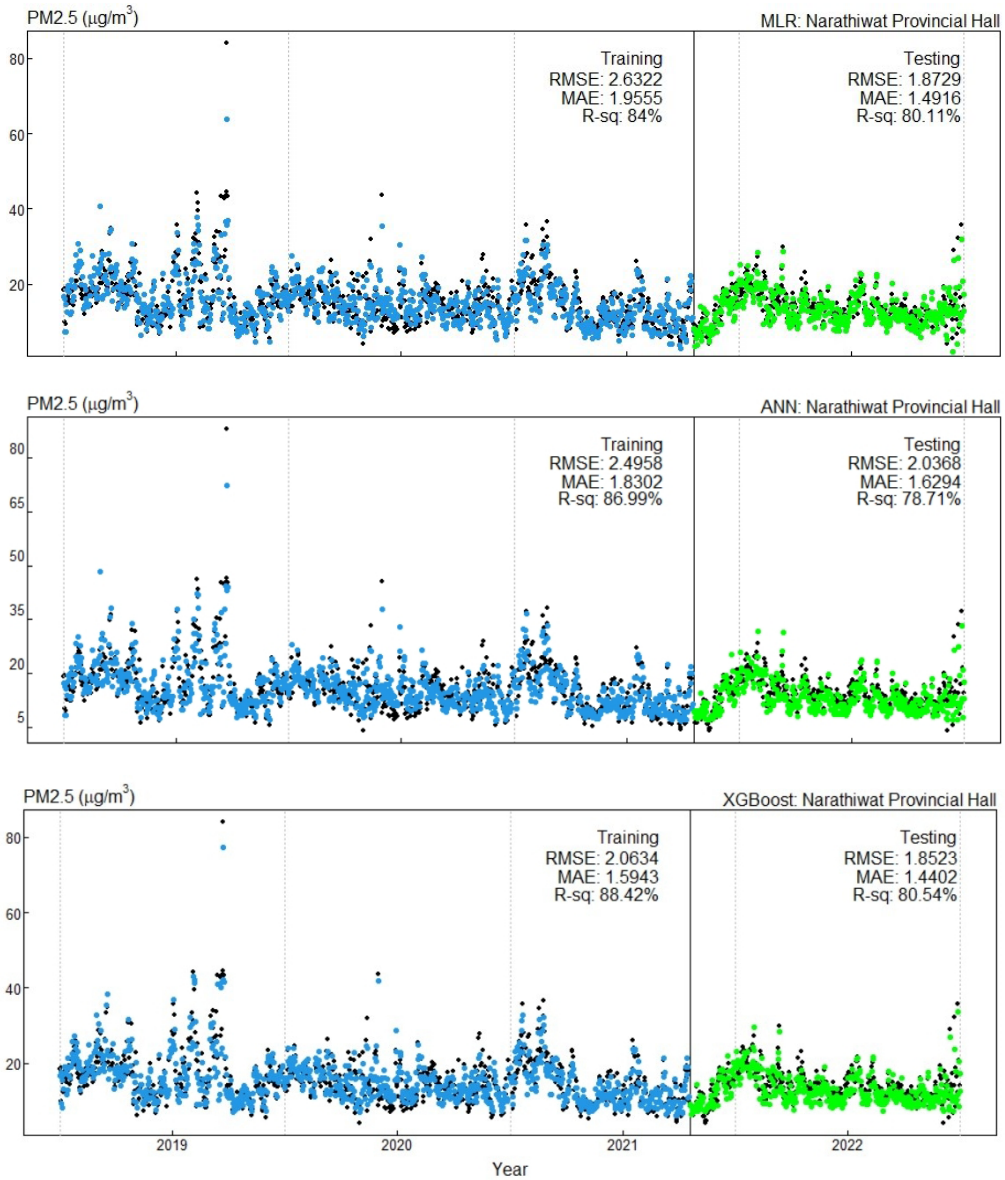

FIGURE 4. The original and fitted PM2.5 trajectories by MLR (top), ANN (middle) and XGBoost (bottom) at the station in Narathiwat

and XGBoost for Narathiwat, while Figure 5 displays the corresponding results for Phuket. Table 5 presents an evaluation of each model's performance at both stations, using metrics such as RMSE, MAE, and R-squared values. Notably, at the Narathiwat station, XGBoost outperformed MLR and ANN on the testing set, achieving RMSE, MAE, and R-squared values of 1.8523, 1.4402, and 80.54%, respectively. Conversely, at the Phuket station, MLR displayed superior performance on the testing set, with MAE and R-squared values of 2.1722 and 61.20%, respectively.

When comparing the model results between the two stations, it becomes evident that the model's performance at Narathiwat surpasses that at Phuket. One potential explanation for this discrepancy is the higher incidence of missing values in the outcome variable (PM2.5) at Phuket, which stands at 12.66%, as opposed to Narathiwat, where it is only 5.82%. This implies that the model tends to perform better when working with more complete datasets. However, in practical scenarios, it is less common to have complete data for natural phenomena. Therefore, a potential solution could involve the use of data imputation methods, and further in-depth research is warranted in this regard.

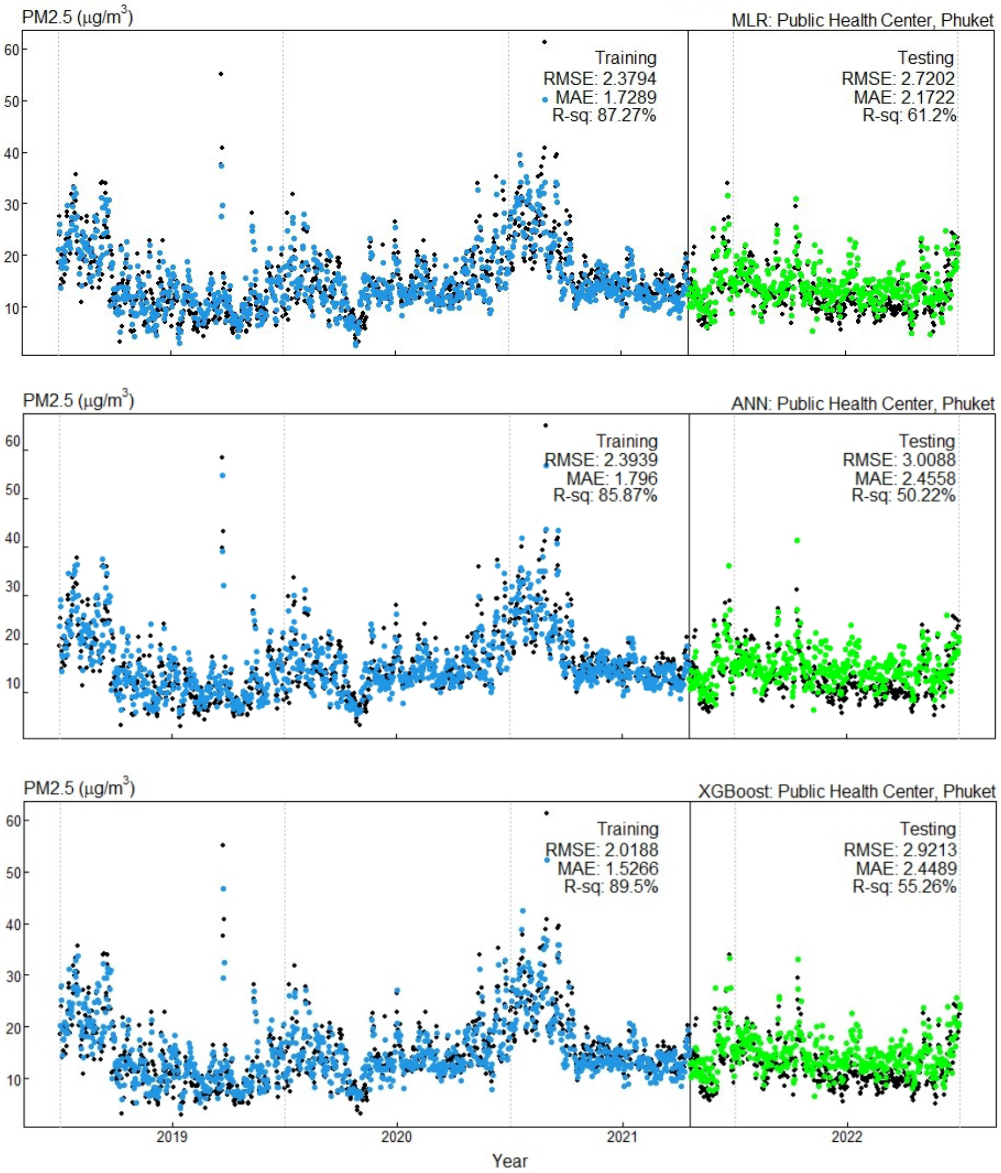

FIGURE 5. The original and fitted PM2.5 trajectories by MLR (top), ANN (middle) and XGBoost (bottom) at the station in Phuket

TABLE 5. Model evaluation for MLR, ANN, and XGBoost for both stations

Stations	**Models**	**RMSE**		**MAE**		**R^2**	
		Train	**Test**	**Train**	**Test**	**Train**	**Test**
Narathiwat	MLR	2.6322	1.8729	1.9555	1.4916	84.00%	80.11%
	ANN	2.4958	2.0368	1.8302	1.6294	86.99%	78.71%
	XGBoost	2.0634	1.8523	1.5943	1.4402	88.42%	80.54%
Phuket	MLR	2.3794	2.7202	1.7289	2.1722	87.27%	61.20%
	ANN	2.3939	3.0088	1.7960	2.4558	85.87%	50.22%
	XGBoost	2.0188	2.9213	1.5266	2.4489	89.50%	55.26%

Focusing on the factors contributing to PM2.5, it turns out that three variables (wind direction, relative humidity, and AOD) at Narathiwat and two variables (NO_2 and AOD) at Phuket did not contribute to PM2.5 concentrations. It was surprising that AOD was not a significant predictor of PM2.5. Nonetheless, the evidence indicates that the statistical modeling results reveal little to no improvement in forecasting the long-term average PM2.5 when AOD is included [17].

4. **Conclusions and Further Study.** A statistical model (MLR) and two machine learning models (ANN and XGBoost) were employed to predict PM2.5 concentrations at two distinct stations in southern Thailand. These models utilized eleven predictors, including four pollutant variables, six meteorological parameters, and one remote sensing dataset (AOD). Different significant predictors at each station can be attributed to local environmental variations, such as geographic and climatic differences. Additionally, variations in data distribution, quality, and sample size, along with differing human activities and socioeconomic conditions, can influence model outcomes. Therefore, a more in-depth study is needed to fully understand and address these factors.

Acknowledgment. This work is partially supported by the National Science, Research and Innovation Fund (NSRF) and Prince of Songkla University (Grant No. SAT6601245S-0), Faculty of Science and Technology, Prince of Songkla University, Pattani Campus, Thailand and Graduate School, Prince of Songkla University, Thailand. The authors also gratefully acknowledge the helpful comments and suggestions of the reviewers, which have improved the presentation.

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